

International Conference on Transforming Education for Future: Education 4.0

Execution Teams

Day: 27th September, 2025 (11:45am-1:15pm)

Track Schedule: AI in Education

Session Chair: Dr. S.B. Nair

Session Co-chair: Dr. Parag Nijhawan

Participants: PPT format uploaded on <https://thec2025.thapar.edu>

Presentation time : 10 min , Q/A :2 min

Venue : TSLAS Room 1 (1st floor TSLAS building)

S. No.	Paper and Authors
1.	Role of Artificial Intelligence in Shaping Cognitive Skill Learning (ID 2) Tanya Garg¹[0000–0002–6915–9023], Raghuveer Singh Dhaka²[0009–0006–0756–2948], Sapanpreet Kaur³, and Vijay Kumar⁴ ^{1,2,3}Thapar Institute of Engineering & Technology, Patiala, Punjab, India, 147004, ⁴Chandigarh Group of Colleges, Jhanjeri, Mohali, 140307 e-mail: tanya.garg@thapar.edu, rdhaka_phd21@thapar.edu, sapanpreet.kaur@thapar.edu, yiju052@gmail.com
2.	Integration of Artificial Intelligence in Flipped Method of Teaching: A Mini Review (ID 26) Rinku Yadav ^a , Amandeep Kaur ^{b*} , Niraj Balac ^{a,b,c} National Institute of Technical Teachers Training and Research, Deemed to be University (Distinct Category) , Sector 26, Chandigarh 160019, India Corresponding Author: Rinku Yadav Email ID: Raoriyanshi11@gmail.com Phone No.: +91-7087491199
3.	Transformative Effect of AI in Education Monika Bhardwaj (ID 34) , Aabhas Bhardwaj² ¹monika_gupta2208@yahoo.com, aabhas.bhardwaj2003@gmail.com ¹Department of Electrical and Electronics Engineering, Maharaja Agrasen Institute of

	Technology, Rohini, New Delhi, India 2Environment, Sustainability, Governance & Tech, Gramener – A Straive Company, Hyderabad, India
4.	Educate, Elevate, Innovate: Integrating Generative AI and Neuromarketing for Enhanced Pedagogical Effectiveness(ID 45) Kawalpreet Singh* Prof. Arvind Kumar a,b,c School of Management, National Institute of Technology Rourkela, Odisha, India a Stephen Young Institute for International Business, University of Strathclyde, Glasgow, United Kingdom b School of Management, National Institute of Technology Rourkela, Rourkela, India c School of Management, Shinawatra University, Pathumthani, Thailand Email - arvind.kumar@strath.ac.uk Contact - +44-7776918614
5.	Role of Artificial Intelligence and Machine Learning in Teaching and Learning(ID 116) <i>Dr. Jyoti Agarwal, Dr. Pradeep Kumar Sharma, Meenakshi Pathak, Anu Saxena Shri Ram Murti Smarak College of Engineering and Technology Bareilly, India jyoti.agarwal@srmscet.edu</i>
6.	Offline Digital Assessment System: Real-Time Quiz Administration and Evaluation Through Local Server Networks(ID 95)- shifted J David Rathnaraj¹, Immanuel R¹, Sanjana N² ¹Mechanical Engineering, Sri Ramakrishna Institute of Technology, Coimbatore, India ²Computer Science and Engineering, Sri Ramakrishna Institute of Technology, Coimbatore, India *Corresponding Author: davidrathnaraj@gmail.com — +91 9876543210

Co Session Chair Duties

1. Arrange one laptop per session
2. Review the session schedule, speaker list, and presentation topics
3. Coordinate with the session chair

4. Communicate with speakers in advance to confirm attendance and presentation needs
5. Ensure all presentation materials (PPT) are submitted and available
6. Assist in introducing speakers
7. Monitor time and help keep presentations within their allotted slots
8. Handle Q&A sessions by collecting audience questions or moderating discussions
9. Maintain decorum and engagement throughout the session
10. Thank speakers and attendees for their participation

Short Bio data of the Session Chair

Prof Shiv Shankar B. Nair

SHIVASHANKAR B. NAIR

PROFESSOR (FORMER HEAD OF DEPT.)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,
INDIAN INSTITUTE TECHNOLOGY GUWAHATI,
GUWAHATI - 781 039 • ASSAM (INDIA)

ASSOCIATE FACULTY,
CENTRE FOR INTELLIGENT CYBER-PHYSICAL SYSTEMS (CICPS) &
CENTRE FOR DRONE TECHNOLOGY, (CDT), IIT GUWAHATI

PHONE: +91 361 2582356(O), 2398(LAB.), 2584356(R)

E-MAIL: sbnair@iitg.ac.in

[HTTP://WWW.IITG.AC.IN/SBNAIR](http://www.iitg.ac.in/sbnair)

[HTTP://WWW.IITG.AC.IN/CSE/ROBOTICS/](http://www.iitg.ac.in/CSE/ROBOTICS/)

Employment (Current)

Professor 2012 - To Date

Dept. of Computer Science & Engineering,
Indian Institute of Technology Guwahati

Associate Faculty

Centre for Intelligent Cyber-Physical Systems &
Centre for Drone Technology,
Indian Institute of Technology Guwahati

Past:

Adjunct Professor 2020

School of Electrical Sciences,
Indian Institute of Technology Bhubaneswar

Professor & Head 2012- 2014

Dept. of Computer Science & Engineering,
Indian Institute of Technology Guwahati, Assam

Associate Professor 2004 - 2012

Dept. of Computer Science & Engineering,
Indian Institute of Technology Guwahati, Assam

Professor - S. Korean Brain Pool (Funded by the South Korean Federation of Science & Technology) 2008 - 2009

HuCare: The Human Centred Advanced Research and Education Centre, Dept. of Instrumentation & Contrc
Hanbat National University,

Chairman, GATE 2007, Indian Institute of Technology Guwahati 2006-2007

Assistant Professor 1998 - 2004

Dept. of Computer Science & Engineering,
Indian Institute of Technology Guwahati, Assam (India)

Senior Lecturer 1986 - 1998

Post Graduate Dept. of Applied Electronics, Amravati University, Amravati, Maharashtra (India)

Visiting Lecturer 1986 - 1988

Department of Electronics & Computer Engineering,
Government College of Engineering, Amravati, Maharashtra (India)

Lecturer 1985 -1986

Department of Electronics Engineering.,
S.S.G.M.College of Engineering, Shegaon, Maharashtra (India)

Education

Ph.D. (Robot-Robot-Human Interactions in Real Time Environments) 1998 Amravati University, Amravati (India)

M.E. (Electronics) 1988 Amravati University, Amravati (India)

M.Sc. (Applied Electronics) 1985 Amravati University, Amravati (India)

Awards

Gold Medal at B.Sc.

First in merit in M.Sc. (Applied Electronics) in the University

First in merit in M.E. (Electronics) in the University

Research Projects

Current: Part of Technology For Underwater Exploration, IIT Guwahati Technology Innovation and Development Foundation (Section 8 Company) (**Rs. 115 crore project**)

Past:

Chiefly responsible for the procurement of funding for the lab. on *Intelligent Robot Human Interaction & Embedded Systems*, at the Department of Computer Sc. & Engg., IIT Guwahati, under the FIST (2009) programme of the DST, Govt. of India.

<http://www.iitg.ac.in/cse/robotics/>

(I) Chakshu: An Assistive User Interface for the Visually Challenged Sponsoring

Agency: Ministry of Social Justice and Empowerment, Government of India

Value of project : INR 10,00,00 (approx.)

Duration : Two years

Status & Designation: Co-PI

(II) A High-End Interface for Robots/Devices Embedded on the Net

http://www.iitg.ernet.in/engfac/sbnair/public_html/iRobIN/robin-index.htm

Sponsoring Agency: Microsoft Corp. USA under the Microsoft Academic Alliance .

Past:

Chiefly responsible for the procurement of funding for the lab. on *Intelligent Robot Human Interaction & Embedded Systems*, at the Department of Computer Sc. & Engg., IIT Guwahati, under the FIST (2009) programme of the DST, Govt. of India.

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(II) *A High-End Interface for Robots/Devices Embedded on the Net*

http://www.iitg.ernet.in/engfac/sbnair/public_html/iRobIN/robin-index.htm

Sponsoring Agency: Microsoft Corp. USA under the Microsoft Academic Alliance .

(III) *Design & Development of An Intelligent Assisting Agent for Microsoft Windows Desktop*

http://www.iitg.ernet.in/engfac/sbnair/public_html/agent_web_page/index.htm

Sponsoring Agency: Microsoft Corp. USA under the Microsoft Academic Alliance

(IV) *Resource Centre for Indian Language Technology Solutions, IIT Guwahati*

Sponsoring Agency: Ministry of Information Technology, Govt. of India.

Consultancy

UNICEF: Real-time monitoring and supportive supervision of VHND and RI using mobile technology

Co-Consultant: The Technical Design of Deep Groove & Roller Bearings

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Execution Teams

Day: 27th September, 2025 (11:45am-1:15pm)

Track Schedule: Faculty Development and Support

Session Chair: Dr. Tanu Tandon

Session Co-chair: Dr. S.C. Bose

Participants: PPT format uploaded on <https://thec2025.thapar.edu>

Presentation time: 10 min, Q/A: 2 min

Venue : Venue : TSLAS Room 3 (1st floor TSLAS building)

S. No.	Paper and Authors
1.	Skill Enhancement Courses Under NEP-2020: A Training Module for Fi Responders to Crime (ID 13) Jasjeet Kaur¹ and Gurvinder Singh Sodhi^{2*} ^{1.} Department of Chemistry, Shaheed Rajguru College of Applied Sciences for Women (University of Delhi) Vasundhara Enclave, Delhi-110096, India ² Forensic Science Unit, Department of Chemistry, S.G.T.B. Khalsa College, University of Delhi, Delhi-110007, India E-mail: gssodhi@sgtbkhalsa.du.ac.in; Phone: 9891157220
2.	Exploring the impact of digitalization on teachers' effective communication(ID 25) Dr Navdeep Dhillon (corresponding author) Assistant Professor School of Humanities and Social Sciences, TIET Patiala, India Email ID- navdeep.dhillon@thapar.edu ORCID- 0000-0001-5622-0051 Dr Sheena Chhabra Assistant Professor Thapar School of Liberal Arts and Sciences, TIET Patiala, India Email ID - sheena.chhabra@thapar.edu
3.	Assessing the pedagogical approach and relationship between personality traits and communication styles of teachers(ID 67) Dr Navdeep Dhillon (corresponding author) Assistant Professor School of Humanities and Social Sciences, TIET Patiala, India Email ID- navdeep.dhillon@thapar.edu ORCID- 0000-0001-5622-0051 Dr Gurvinder Kaur

	Associate Dean Outreach School of Humanities and Social Sciences Thapar Institute of Engineering and Technology, Patiala India Email ID- gurvinder@thapar.edu
4.	Reflective Teaching as a Practical Approach to achieve Academic Excellence(ID 82) <i>Parag Nijhawan</i> Associate Professor Department of Electrical and Instrumentation Engineering Thapar Institute of Engineering and Technology, Patiala (PUNJAB), INDIA parag@thapar.edu
5.	Role of Appropriate and Timely Assessment and Feedback in Higher Education(ID 94) Manjeet Singh^a, Parag Nijhawan^a, Sunil Kumar Singla^a manjeetsingh.tietphd@gmail.com, parag@thapar.edu, ssingla@thapar.edu ^a Electrical and Instrumentation Engineering Department Thapar Institute of Engineering and Technology, Patiala (PUNJAB), INDIA
6.	Financial Literacy as a Protective Factor Against Materialism-Driven Impulsive Buying(ID 148) Bhawna Sehgal, Ph.D Scholar, Thapar School of Liberal Arts and Sciences (TSLAS), Thapar Institute of Engineering & Technology, Patiala, bsehgal_phd24@thapar.edu, +91 8091306820 Dr. Amanpreet Kaur, Assistant Professor, Thapar School of Liberal Arts and Sciences (TSLAS), Thapar Institute of Engineering & Technology, Patiala, amanpreet.kaur1@thapar.edu, +91 8054200162
7.	Deep Learning Framework for EEG-Based Mind Wandering Detection in Education (ID 39) – Shifted to Faculty development and support 1Mehar, Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, Patiala, India, mmehar_be23@thapar.edu Sandeep Mandia, Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, Patiala, India, smandia20@gmail.com 3Anil Arora, Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, Patiala, India, anil.arora@thapar.edu 4Kulbir Singh, Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, Patiala, India, ksingh@thapar.edu *Corresponding Author: smandia20@gmail.com

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Bio data of Session Chair

Prof. (Dr.) Tanu Tandon is the Director of the University Institute of Teachers Training and Research (UITTR) at Chandigarh University, and brings with her over 23 years of rich experience in teacher education, academic leadership, and institutional development. She is a Chancellor's Gold Medalist from Mumbai University and was awarded her Ph.D. from Lucknow University by the Indian Council of Social Science Research (ICSSR). She is also an IIM Lucknow Alumnus. Throughout her distinguished career, she has established Departments of Education, Faculty Development Centres, and Centres for Women and Gender Studies across several prestigious universities in India. She has mentored more than 15,000 teachers and lecturers, conducted numerous national and international workshops, and led faculty training programs with a deep commitment to inclusivity, academic innovation, and reflective teaching. Prof. Tandon has also represented Teacher Education at a national consultative meet at NITI Aayog, contributing valuable insights to education policy and practice at the highest level. At Chandigarh University, she leads UITTR with a clear mission—to empower educators not just as instructors, but as mentors, motivators, and agents of change."

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Day: 27th September, 2025 (11:45am-1:15pm)

Track Schedule: Global Implications of Education and Academia Industry connect

Session Chair: Dr. Van Hans

Session Co-chair: Dr. Karminder

Participants: PPT format uploaded on <https://thec2025.thapar.edu>

Presentation time: 10 min, Q/A: 2 min

Venue : TSLAS Room 2 (1st floor TSLAS building)

S. No.	Paper and Authors
1.	Breaking Cultural Barriers in Education: MOSAIC's Multimodal AI for Cross-Cultural Learning and Heritage Preservation (ID 130) Madhav Arora ^{1*} , Bani ² and Neeru Jindal ³ ¹ *Electronics & Computer Engineering, TIET, Patiala, India. ² Computer Engineering, TIET, Patiala, India. ³ Department of Electronics and Communication Engineering, TIET, Patiala, India. neeru.jindal@thapar.edu
2.	Reimagining Engineering Education in India: A Strategic Framework for Quality Enhancement (ID 48) Alka Pandita and Ravi Kiran Thapar Institute of Engineering and Technology, Patiala 147004, India apandita@thapar.edu
3.	AI in Mathematics Education: A Critical Review of Its Role, Impact, and Pedagogical Implications in Higher Education (ID 40) Kavita Goyal ^{*,a} ^a Department of Mathematics, TIET Patiala, India *kavita@thapar.edu
4.	‘Valley of Death’: A Case for Bridging the Gap for Trans-Disciplinary (ID 56) Amir Raza, Dr Meenakshi Sood, Dr Rajesh Mehra

	amir.cdc25@nitttrchd.ac.in; meenakshi@nitttrchd.ac.in, rajeshmehra@nitttrchd.ac.in National Institute of Technical Teachers Training and Research, Chandigarh, India Amir Raza; E: amir.cdc25@nitttr.ac.in
5.	From Classroom to Cubicle: Decoding the Internship Experience as a Career Catalyst(ID 144) <i>Bani , Madhav Arora, Neeru Jindal</i> <i>Electronics & Computer Engineering</i> <i>Department of Electronics and Communication Engineering</i> bbani_be24@thapar.edu
6.	Project-based Learning and Employability: A Cognitive Constructivism and Transformative Learning Perspective from a Developing Country (ID 172) Dr Karminder Ghuman, Dr Sakshi Bathla, Dr Saurabh Mittal LM Thapar School of Management, TIET (Deemed to be University), Patiala karminder@thapar.edu

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Short Bio Data of Session Chair :

Dr. Hans van Ees obtained his PhD in economics. He is Emeritus Professor Corporate Governance and Institutions and Founding Dean of the University College Groningen, University of Groningen, The Netherlands. He served in several visiting positions outside the Netherlands and is currently associated with Thapar Institute of Engineering and Technology, India as professor and honorary dean of research at the School of Liberal Arts and Sciences. Furthermore, he is a regular member of national and international accreditation committees for academic /programs in Economics and Business. and a Fellow of the European Academy of Management. In his research Hans van Ees studies the complexities of corporate governance and the interdependency between corporate governance and higher order institutions in a multicountry perspective. He served on the editorial board of several academic journals in corporate governance, published research and supervised PhD projects on corporate governance, board of directors and stewardship. In addition he is involved in consultancy and research for private and public organizations on the principles of good governance.